

Mid-Semester of 3rd Sem B.Tech.

MID-SEMESTER EXAMINATION, JUNE 2026 (SUMMER SEMESTER)

COURSE CODE: EPMTTC301

COURSE TITLE: MATHEMATICS FOR MACHINE LEARNING

TIME: 1.30 HRS

MAX. MARK: 20

Note:- Attempt all questions. Missing data/ information (if any) may be suitably assumed & mentioned in the answer.

Q. NO	QUESTION	MARKS	CO
1a	$T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ defined by $T(x, y, z) = (y - z, y + z)$. Determine the matrix of the linear transformation T with respect to the standard ordered basis.	2	CO1
1b	Examine if the set $W = \{(x, y, z, w) \in \mathbb{R}^4 : z = 100\}$ is a subspace of $\mathbb{R}^4$.	2	CO1
2a	Check whether $S = \{(1, 2, 3), (1, 1, 1), (1, 0, 1)\}$ of $\mathbb{R}^3$ is linearly independent or linearly dependent	2	CO1
2b	Find $\text{Ker}T$ of the linear mapping $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ defined by $T(x, y, z) = (x + y, y - z)$.	2	CO1
3a	Examine whether T is a linear mapping where $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(x, y) = (x - 3y, 2x + y)$.	2	CO1
3b	Let $u = (1, 2), v = (3, 1) \in \mathbb{R}^2$. Find $w \in \mathbb{R}^2$ such that $\langle w, u \rangle = 4$, $\langle w, v \rangle = 5$, where $\langle \cdot, \cdot \rangle$ is the standard inner product on $\mathbb{R}^2$.	2	CO2
4a	Check whether the quadratic form $Q(x, y, z) = 2x^2 + 3y^2 + 5z^2 + 2xy - 2xz + 4yz$ is positive definite or positive semi-definite.	2	CO2
4b	Find Eigen values of the matrix $A = \begin{bmatrix} 2 & 1 & 2 \\ 1 & 2 & 2 \\ 2 & 2 & 2 \end{bmatrix}$.	2	CO2
5a	Use Gram-Schmidt process to convert the basis $\{(1, 0, 1), (1, 1, 0), (0, 1, 1)\}$ of the Euclidean space $\mathbb{R}^3$ with standard inner product into an orthogonal basis.	2	CO2
5b	Show that $S = \{(1, 2, 2), (2, -1, 0), (2, 2, -3)\}$ is an orthogonal basis of $\mathbb{R}^3$. Hence, find an orthonormal basis of $\mathbb{R}^3$.	2	CO2